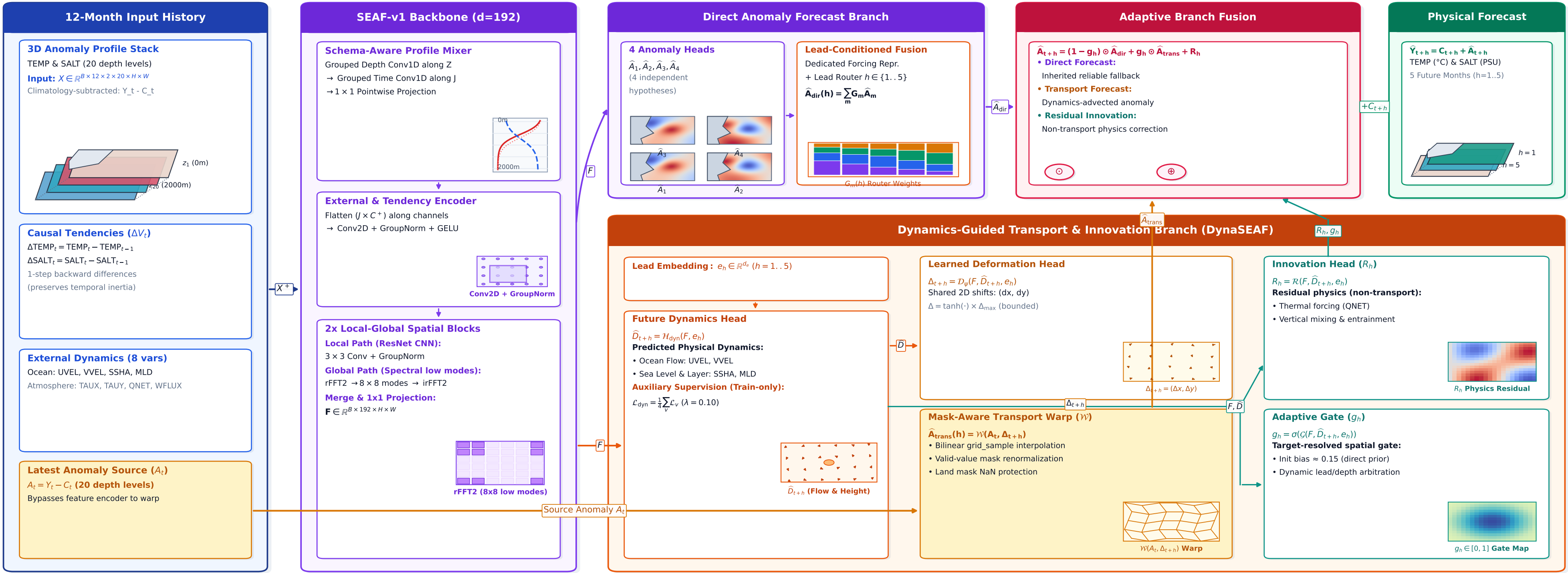


DynaSEAF: Dynamics-Guided Transport & Innovation for Ocean Anomaly Forecasting

Dual-branch architecture combining inherited direct anomaly forecasting with dynamics-conditioned effective transport, residual innovation, and adaptive gating



Training-Only Auxiliary Supervision: $\mathcal{L}_{\text{dyn}}$ backpropagates through $\mathcal{H}_{\text{dyn}}$ and backbone F (labels absent during validation/test)

External Persistence-Aware Evaluation Protocol (Independent Baselines - Not Model Inputs)

